

JÖNKÖPING UNIVERSITY
*School of Engineering*

# Designing Data Visualizations for ADHD Users

## Effects on Satisfaction and Cognitive Load

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## Abstract

Attention Deficit/Hyperactivity Disorder (ADHD) can significantly influence how individuals process information and interact with digital interfaces. While cognitive load has been studied in user experience research, the specific needs of ADHD users remain underexplored. This study investigates how data visualization strategies within recommender systems affect cognitive load and user satisfaction among individuals with ADHD compared to neurotypicals.

Two high-fidelity movie recommender system prototypes were developed in Figma: one utilizing progressive disclosure with additional contextual information (e.g., tags, genre, ratings, and duration), and another presenting minimal information. Usability testing was conducted with both ADHD and neurotypical participants. Cognitive load was measured using the NASA-TLX, while user satisfaction was assessed using the USE Questionnaire. Semi-structured interviews were conducted to complement the quantitative findings with qualitative insights.

The results indicate that participants with ADHD experienced significantly lower cognitive load and higher satisfaction when using the prototype that provided additional information through progressive disclosure. In contrast, the minimal-data interface increased perceived workload and reduced satisfaction among ADHD users. Neurotypical users also preferred the prototype providing more information, although the effects were less pronounced. Qualitative data revealed that ADHD users relied heavily on information such as ratings, genres, tags, and running time when making decisions and consistently expressed a desire for greater transparency and access to information.

The findings suggest that accessible data visualization can support decision-making and reduce cognitive load for users with ADHD. Future research should examine alternative data visualization and interface design strategies, investigate visual attention through eye-tracking methods, and explore how these findings transfer to other digital environments, such as e-commerce platforms, where users must navigate large amounts of information and make complex decisions.

# 1 Introduction

As digital platforms continue to rely on recommender systems to assist users in finding relevant content, understanding how information presentation affects the user experience becomes increasingly important. Different users may rely on different types of information when making a choice, which raises questions about how interfaces can be designed to support a wide range of needs.

According to Cognitive Load Theory, working memory has a limited capacity for processing information. When the amount of information that must be processed exceeds this capacity, individuals may experience cognitive load, making it more difficult to efficiently process information and complete tasks (Sweller et al., 2011).

This study focuses on neurodivergent individuals, specifically people with Attention-Deficit Hyperactivity Disorder (ADHD). According to the American Psychiatric Association (2013), ADHD is a neurodevelopmental disorder that is characterized by inattention, inability to wait, and disorganization. People with ADHD tend to overlook important information when doing a task due to a reduced level of attention, as well as a tendency to forget about the task itself. In this paper, we argue that for web interfaces to be accessible, they need to consider neurodivergent individuals and their specific needs, as these user groups have a different way of understanding and processing information (Wu et al., 2021).

These challenges may become particularly relevant in recommender systems, where users are often required to process multiple sources of information. This may affect cognitive load, potentially making decision-making more demanding for users with ADHD. Therefore, understanding how information can be presented in a more accessible manner is an important design consideration.

To accommodate their needs, data visualization techniques have been proposed, including an existing set of visualization design guidelines by Wu et al. (2021) alongside the interactive visualization technique proposed by Richthammer et al. (2017), in which they introduced an interactive visualization approach where movies are presented to users with supplementary data, using graphs and text to aid decision-making was tested. The visualization design guidelines by Wu et al. (2021) specify the types of graphs that are accessible for neurodivergent users. Building on this, this study investigates whether the presentation of information in a movie recommender system can reduce cognitive load and improve user satisfaction for individuals with ADHD.

Focusing on how cognitive load is influenced by different ways of displaying data, two prototypes were tested that feature a movie recommender system interface. In this

context, the recommender system serves as a personalized filtering tool that suggests content based on user preferences, which can require significant cognitive processing to navigate.

To measure cognitive load, the NASA-TLX scale was used (Cao et al., 2009) and the USE scale was used to measure user satisfaction (Lund, 2001). The NASA-TLX measures cognitive load by asking users how hard they had to work (Cao et al, 2009), while the USE scale is a simple questionnaire that measures Usefulness, Satisfaction, and Ease of Use (Lund, 2001).

Based on the results, this research aims to propose design solutions that help reduce cognitive load, increase the satisfaction of neurodivergent users in their daily lives, and encourage a more inclusive approach to be the default (Chattopadhyay, 2025).

## 1.1 Problem statement

Cognitive Load in web interfaces has been studied for a long time, yet only a few articles have studied ADHD individuals. Up until now, the main focus has been to make the interfaces as intuitive as possible for the general population, often assuming neurotypical patterns of attention, memory, and information processing.

This focus creates an "epistemic imbalance" (Spiel et al., 2022). In the context of ADHD, technology research is frequently shaped by goals that value neuro-normative outcomes. Instead of designing to support the unique ways people with ADHD navigate information, technology is often used to "mitigate" behaviors that are seen as disruptive to neurotypical standards. This approach treats ADHD as a clinical problem to be managed rather than recognizing the user as an expert of their own lived experience. Instead of designing for the person, the research focuses on fixing behaviors that do not fit a neurotypical mold.

Furthermore, traditional research methods often fail to properly engage people with ADHD during the early design stages (Kalantari et al., 2025). This is often due to a reliance on what Ekstrand and Willemsen (2016) call "implicit feedback", which is data collected by simply watching visible user behavior because it is easy to collect and good at predicting actions. However, this approach has flaws. Researchers and users have different visions for what technology can do. For example, Moroyoqui et al. (2022) point out that while designers focus on technical efficiency, users with ADHD often find highly restrictive interfaces to be a barrier to their autonomy. Users may perceive rigid system constraints as an attempt to micromanage their behavior, which may lead to frustration and a lack of engagement with the tool. As Moroyoqui et al. (2022) explain, technology for ADHD is most successful when it supports "spontaneous" daily life rather than forcing a person into a strict, narrow system.

Since many studies rely on neurotypical interview styles, they miss the "insider perspective" of those who actually live with ADHD. Because of this, many web interfaces still have a "deficit frame", in which the user is perceived as having an issue that has to be resolved rather than the interface being seen as a tool that is failing to be inclusive.

Beyond the general layout of web interfaces or applications, a critical barrier exists in how information itself is visualized. While data visualization is intended to simplify complex concepts, most design guidelines are built exclusively for "non-disabled" populations, as noted by Wu et al. (2021). This creates a real barrier in how people actually use and understand technology. Instead of helping users understand their options, graphs and layouts can feel overwhelming and confusing.

As Wu et al. (2021) highlight, making data accessible is a fundamental requirement for self-determination. When interfaces fail to provide clear, intuitive visualizations, they effectively take away the user's power to make informed, data-driven decisions.

While the specific cognitive needs of different populations vary, research into inclusive design provides a vital parallel for this study. For example, Wu et al. (2021) examined how individuals with intellectual and developmental disabilities (IDD) perceive various graph types compared to neurotypical users. Their findings revealed that while design preferences were often similar across groups, the impact of those designs were significantly more profound for the IDD group.

Other studies that have explored the impact of data visualization within recommender systems found that users were very satisfied with having more information about their recommendations, and have shown an increased trust due to the transparency of the data (Kunkel et al., 2017; Richthammer et al., 2017).

Furthering the collective work, this research will combine proposed design guidelines in an attempt to create a more accessible and enjoyable user experience for people with ADHD, who have largely been disregarded in the development of apps and web interfaces. Recommender systems are beneficial for managing large amounts of information, and are commonly integrated into apps used in our daily lives. For this reason, a recommender system interface was tested.

## 1.2 Purpose and research questions

Drawing on the problem statement, it is evident that standard web interfaces often prioritize a "one size fits all" design that may not align with diverse neurological processing styles. Further, it is clear that when design guidelines ignore these differences, they create barriers to independence and engagement (Wu et al., 2021).

Consequently, the purpose of this study was to examine how different types of data visualization, specifically within the context of streaming-based recommender systems, influence the cognitive load and satisfaction of users with ADHD.

To fulfill this purpose, the following research questions have been defined:

**RQ1: How does data visualization affect the cognitive load of users with ADHD in recommender systems?**
While visualization is meant to simplify data, it could also create a "sensory noise". This question examines whether this tool, in the form of progressive disclosure, actually lightens the cognitive load for neurodivergent users or if they add an extra layer of stress when making decisions (Wu et al. 2021).

**RQ2: Are there differences between ADHD and neurotypical users in how data visualization influences cognitive load?**
To build truly effective tools, current design standards must be investigated to decide if they work for everyone or if they were only made with one type of user in mind. Progressive disclosure acts as a solution to "choice overload" by hiding complex details until they are actually needed, and it allows the user to focus on one decision at a time.

By comparing how users navigate a movie recommendation interface that shows all data at once versus one that unfolds information more gradually, we can see which designs improve cognitive accessibility for ADHD users and which ones cause unnecessary stress.

**RQ3: How does data visualization influence user satisfaction among users with ADHD?**
This question explores whether visual aids make neurodivergent individuals feel more confident and in control, or if the design causes frustration and exclusion.

## 1.3 Scope and delimitations

In this study, we wanted to test how using data visualization within recommender systems affects the cognitive load and user satisfaction for individuals with ADHD.

Two groups were recruited when conducting this study: one consisting of users with ADHD and a control group consisting of neurotypicals.

The variables that measured were cognitive load and user satisfaction for both groups, and then they were compared against each other. The independent variables in this study were the different types of data visualization, and our dependent variables were cognitive load and user satisfaction.

The study was conducted both online and in-person settings, with participants tested individually. The target audience for this study was young adults with a diagnosis of ADHD who have previously used recommender systems and have at least a moderate level of digital literacy.

Participation was made on a trust-based system, as the interviewees were asked if they were diagnosed to be part of the test, but due to ethical concerns and data privacy, there was no medical evidence of the diagnosis.

Additionally, the prototype tested was made using Figma, a collaborative platform for designing web interfaces, instead of a personalized recommender system.

## 1.4 Disposition

This paper is organized into several chapters that guide the reader from the initial problem to the final conclusions. The structure is as follows:

- Abstract
- Chapter 1: Introduction - This chapter sets the stage by presenting the problem statement, which highlights the design gap for users with ADHD in digital interfaces. It also defines the study's purpose and the specific research questions regarding cognitive load and user satisfaction.
- Chapter 2: Theoretical Framework - This section explores the costing literature and core concepts that ground the study. It discusses cognitive load theory, information overload, and the principles of information visualization. It provides the academic foundation that is needed to understand the experimental results.
- Chapter 3: Method and implementation - The work process and experimental design are described and motivated. This includes details on the movie recommendation prototypes, the recruitment of participants with ADHD, and the methods used for data collection and analysis, such as the NASA-TLX and USE scale, and semi-structured interviews.
- Chapter 4: Results - This chapter presents the findings gathered from the usability testing and semi-structured interviews. The quantitative results from the NASA-TLX and USE questionnaires are analyzed to compare cognitive load and user satisfaction between ADHD and neurotypical participants across two prototype conditions. In addition, the chapter presents the qualitative findings obtained through thematic analysis, highlighting participants' experiences, decision-making strategies, and perceptions of the different interface designs.

- Chapter 5: Discussion - This chapter discusses the results in relation to the study's research questions, theoretical framework, and previous research. The findings are interpreted to examine how data visualization and progressive disclosure influence cognitive load and user satisfaction among users with ADHD compared to neurotypical users. The chapter also reflects on the strengths and limitations of the study and evaluates the validity and reliability of the research process.
- Chapter 6: Conclusions and further research - This chapter summarizes the main findings and conclusions of the study. It discusses the implications of the results for inclusive and accessible design, particularly for users with ADHD. The chapter also identifies limitations of the current study and proposes directions for future research, including the investigation of alternative visualization strategies, eye-tracking methods, and applications in other digital contexts such as e-commerce platforms.

# 2 Theoretical framework

## 2.1 Cognitive load

The Cognitive Load Theory states that the human working memory is divided into three categories: intrinsic, extraneous and germane. While intrinsic load is determined by the information presented itself, the extraneous load is determined by how the information is presented and is often unnecessary. The germane load refers to the working memory resources required when processing, learning and automating information to long-term memory. It facilitates pattern recognition, thus lowering the mental resources used when faced with similar looking information. When processing information, all types add up when consuming working memory resources, and the moment it exceeds the capacity, the cognitive system will fail to understand. When the extraneous cognitive load is too high, due to poor data arrangement, the intrinsic cognitive load cannot make sense of the information anymore. This may lead to the individuals giving up on the information entirely (Sweller et al., 2011).

In the context of web interfaces, cognitive load is very important to be taken into consideration when designing for users with diverse processing styles. For neurodivergent individuals, such as those with ADHD, the threshold for where information becomes overwhelming can be different than that of neurotypical users. As noted by Wu et al. (2021), this user group often interacts with data in unique ways. For instance, it is possible that for neurodivergent individuals, traditional and busy interfaces could act as a processing barrier, making it more difficult to extract relevant information effectively.

## 2.2 Progressive disclosure

According to Yablonski (2020), progressive disclosure is a design strategy used to manage how much information a user sees at any given time. Instead of overwhelming the user with all details at once, the interface defaults to showing only the most essential content. Additional layers of information remain easily accessible but stay hidden until the user actively chooses to see them. In practice, this technique uses familiar elements like dropdown menus or “read more” toggles to tuck away secondary details. By deferring complex data to these secondary layers, the interface becomes more streamlined. For the user, this means less visual clutter and a more focused experience. This is a powerful tool because it directly reduces the cognitive load, making it easier for a person to stay focused on their primary goal, such as choosing a movie (Yablonski, 2020).

This technique is the primary variable being tested in this study. While Richthammer et al. (2017) suggest that reducing visual complexity is beneficial for accessibility, we

propose that progressive disclosure may be specifically powerful for neurodivergent users. This study tests whether unfolding information directly lowers the cognitive load compared to a traditional "all-at-once" layout.

### 2.3 Information overload

While researchers have extensively explored this concept, Belabbes et al. (2023) point out that there is no agreed-upon definition of what information overload is. As cited by Belabbes et al. (2023), one popular definition is by Bawden and Robinson (2009) who describe information overload as a state where a person's "mental bandwidth" is exceeded, leading to a loss of control over the information they are trying to process.

### 2.4 Choice overload

Another related concept is choice overload. When a user is faced with a choice, the time they take to make the decision is relative to the number and complexity of the options (Yablonski, 2024). The decision process increases the mental effort of the users, directly impacting the cognitive load. Moreover, having more choices can lead to user dissatisfaction (Iyengar & Lepper, 2000). In web interfaces, such as recommender systems, understanding choice overload is essential, as their design can easily disrupt the user experience. As suggested by Yablonski (2024), the oversimplification of complex systems is one of the most important tasks of designers.

### 2.5 Data visualization

Data visualization is a tool meant to help the human brain find patterns in a "sea" of information. While we often hear that "a picture is worth a thousand words," Pfitzner et al. (2003) argue that is only true if the picture is actually easy to understand.

### 2.6 Literature Review

Our search strategy was to include all the theories that are going to be included, as well as broaden our search from just ADHD to other neurodivergencies and cognitive disabilities.

The search was structured using specific keywords to find the intersection of user psychology, interface design, and assistive technology. By using "AND/OR", we were able to narrow down thousands of articles to find the ones most relevant to our research.

**User Experience:** We looked for terms like "cognitive load", "choice overload"," information overload", and "decision fatigue" to identify the specific points where the processing effort becomes too high for a user to navigate comfortably.

**Design solutions:** Keywords such as "progressive disclosure" and "recommender systems" were used to find technical ways to manage the processing effort.

**Context:** To keep the results relevant to our study, every search required the inclusion of “neurodivergent” alongside “web”, ”interfaces”, or “apps”.

**Search query:**

[All: "cognitive load"] OR [All: "choice overload"] OR [All: "information overload"] OR [All: "decision fatigue"] OR [All: "progressive disclosure"] OR [All: "recommender systems"] AND [All: "neurodivergent"] AND [[All: "web"] OR [All: "interfaces"] OR [All: "app"]]

Overall, most of the research that included ADHD participants, were not focusing on them individually, but rather as part of a group. In the study made by Dai et al. (2023) for example, the researchers have paired up neurodivergencies with people that experience cognitive decline, to investigate the accessibility features in the online banking system, which were found to be lacking. Studies such as the one conducted by Wu et al. (2021) have studied individuals with Intellectual and Developmental Disabilities, which include ADHD as a neurodevelopmental disorder, in the context of data visualization.

The literature review made by Spiel et al. (2022) highlights this absence of studies focused on ADHD individuals and the lack of technology designed to accommodate these individuals rather than correcting their behaviour.

In our search, two studies were found which proposed a technology specifically for individuals with ADHD. Firstly, Moroyoqui et al. (2022) created an app that focuses on reducing the cognitive load and stimuli that make daily life actions difficult. Secondly, Tang et al. (2021) proposes a graphical interface designed to keep the users engaged and minimize the risks of accidents when driving.

For a more detailed overview of the studies, see Appendix 5.

# 3 Method and implementation

To recruit participants, the university student community and extended peer networks were contacted. Two high-fidelity prototypes were developed using Figma. Both of them function as movie-streaming interfaces, but they differ in how they present information to the user (Figure 1 and Figure 2). Our initial plan was to recruit 20 participants and split equally between neurotypical individuals and those diagnosed with ADHD. In an effort to increase our data accuracy, extra participants were ultimately brought in for the initial testing phase. However, due to last-minute changes in participant availability, the final groups did not end-up perfectly equal. For the interface with minimal data, 6 participants with ADHD and 8 neurotypical individuals were tested. For the progressive disclosure interface with extra data, the

groups consisted of 5 participants from the ADHD group and 5 from the neurotypical group.

Prior to the experiment, a pre-briefing was conducted to make sure the participants were aware of the study objectives and their expected tasks. We emphasized that participation was entirely voluntary and that they had the right to withdraw at any stage without any consequences. After obtaining formal informed consent, participants' screens were recorded during the session to capture their interaction with the movie-streaming prototypes for our analysis. These recordings served as a vital tool for our analysis, as it allowed us to measure the task completion time and identify moments of hesitation.

The experimental session began with free exploration of the interface so that the participants could familiarize themselves with the layout. Following this, two distinct directed-choice scenarios were presented: first, selecting a film based purely on their own personal interest, and second, choosing a movie appropriate for a family viewing setting. To conclude the session, a semi-structured interview was conducted to uncover the "why" behind the participants' actions. Primary motives were identified for each selection and gathering their feedback on the interface's utility. Specifically, we questioned whether the level of data provided was helpful, asking the participants to reflect on whether they required more transparency or a more "quiet" display to make their final decision.

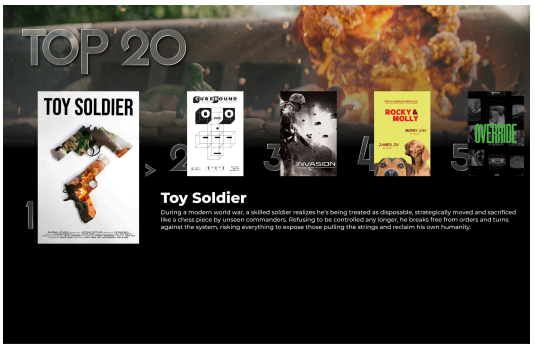



Figure 1. Minimal data

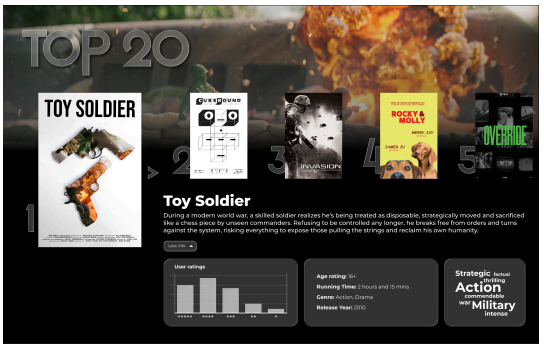



Figure 2. Extra data

After the participants were done with the prototype, they were asked to complete the NASA-TLX questionnaire to measure the cognitive load and the USE Questionnaire proposed by Lund (2001) to measure the user satisfaction. They were also interviewed with open-ended questions about their user experience.

To evaluate the user satisfaction and measure cognitive load, both the NASA-TLX questionnaire and the USE questionnaire were used, as well as a semi-structured interview. The questionnaires were chosen due to their easy-to-use nature and small amount of resources they require, and the semi-structured interviews to complement the quantitative data with more in-depth qualitative data. The specific set of questions

in Appendix 3 and the data collection templates used during these sessions are provided in Appendix 2.

Wu et al. (2021) propose a set of visualization design guidelines for individuals with intellectual and developmental disabilities. They tested different types of graphs on neurodivergents and neurotypicals. Their results found that improving data accessibility for neurodivergents does not bother neurotypicals, urging for a push towards implementing the guidelines to the norm.

While general accessibility guidelines are a great start, we also needed to look at the specific challenge of recommender systems. According to Richthammer et al. (2017), the way data is visualized is just as important as the "math" behind the recommendations. They argue that transparency is key, as users feel more satisfied when they understand why a certain movie is being suggested to them. Furthermore, providing interactive control by giving users the ability to filter and sort their own data helps them feel more in charge of the process. Finally, they emphasize the importance of reducing information noise, noting that in recommender systems, it is easier for a user to feel "lost" in the options. Therefore, the paper suggests using visual aids to help people navigate those choices without feeling overwhelmed.

Using those three sets of guidelines, we strive to find the best way to design interfaces that actually support how different minds work. Ultimately, we hoped to show that when we design for neurodivergence, we create a more satisfying experience for everyone.

## 3.1 Data collection

**RQ1: Measuring Cognitive Load**
To understand how visualization affects the cognitive load, the NASA TLX was used. This is a validated tool that lets participants rate their perceived workload across six dimensions, such as mental demand and frustration. By using this quantitative scale immediately after they finished the movie selection task, an accurate "snapshot" of the cognitive load was obtained.

**RQ2: Comparing User Groups**
To see if design choices like "progressive disclosure" act as a bridge or a barrier for the different user groups, the two different groups (ADHD and neurotypicals) were compared. Both groups used the same prototypes, so the results could be statistically compared to see if certain designs specifically help or hinder users.

**RQ3: Evaluating User Satisfaction**
To explore whether visual aids make users feel more satisfied, the USE questionnaire and semi-structured interviews were used. These conversations took place right after

the tasks were finished. Unlike a strict survey, this format allowed us to hear the insider perspective of both users with ADHD and neurotypical users, letting them explain in their own words what felt satisfying or frustrating about the interface.

## 3.2 The experiment

The experiment was conducted using two custom-built prototypes in Figma that mimicked a movie streaming website. To ensure the results are based on the design variations, original movie posters and summaries were created so that participants were not influenced by movies they already know.

Each session followed a standardized sequence:

1. Introduction: Explanation of the task and obtaining informed consent.
2. The tasks: Participants were asked to find and "pick" a movie.
3. Measurement: Administering the NASA TLX and USE questionnaires.
4. Discussion: Semi-structured interview to capture the feedback from the participant's experience.

## 3.3 Data analysis

Because this study collected different types of information, the analysis was divided into two parts to ensure accuracy. By looking at both the quantitative statistics and the qualitative reflections, we aimed to build a more complete picture of how the visualization affected the participants.

**Quantitative analysis**

For the first part of our analysis, the focus will be on the numerical data gathered from the NASA TLX score and USE questionnaires. The procedure was as follows:

- Compare the groups: statistical methods like independent sample t-tests were used to compare the results between groups.
- Testing the design: investigating differences in cognitive load between the "progressive disclosure" prototype and the "all-at-once" version to see if the design actually reduced cognitive load.

**Qualitative analysis**

Numbers can tell us what happened, but the qualitative analysis tells us more about why. The transcripts from the semi-structured interviews were analyzed using a thematic approach (Braun and Clarke, 2006). The thematic analysis focused on the following aspects:

- Finding patterns: We carefully read through the interview answers to find recurring themes, such as specific design elements that cause frustration or moments where the user felt a sense of ease and satisfaction with their choices.
- The personal perspective: This part of the analysis was dedicated to respecting the lived experience of our participants. We looked for insights that the

numbers might have missed, such as how the extra data affected their confidence or sense of control.

## 3.4 Validity and reliability

For this study's findings to be meaningful, the results must be characterized by both accuracy and transparency in the data collection and analysis phases. Validity was ensured by selecting evaluation frameworks specifically designed to capture the nuances of ADHD interaction, primarily through the NASA TLX and USE scales. Furthermore, to maintain high content validity, original movie titles and summaries were created specifically for this research. This was to ensure that the participants' choices were not based on familiarity.

Reliability means that if another researcher followed our steps, they would reach the same conclusions. By following a standardized procedure where every participant was presented with the exact same instructions and completed the same set of tasks, we ensured consistent results with minimal influence by the researchers. Furthermore, transparency was prioritized, by documenting every part of our recruitment, the design guidelines used, and our analysis methods clearly. This approach allows others to verify our work and ensures the study is built on a repeatable foundation. To strengthen this further, data triangulation was utilized Cartel et al. (2014), combining quantitative NASA-TLX and USE data with qualitative semi-structured interviews.

## 3.5 Considerations

In this study, ethical and social considerations were prioritized to ensure that the research is conducted with respect for all participants.

**Ethical Data Management and Privacy**

Protecting the privacy of participants was a primary responsibility throughout this study. As the research involved individuals with ADHD, a trust-based requirement approach was used, and participants were not required to provide medical evidence of a diagnosis in order to protect their sensitive health information. Furthermore, all data collected during the study was anonymized immediately, and no names or other identifying information were linked to the results. This ensured that participants could not be identified in the final report and that their confidentiality was maintained throughout the research process.

**Informed consent**

Participation in the study was entirely voluntary, and informed consent was obtained from all participants before the study began. Prior to completing any tasks, participants received a clear explanation of the study's purpose, procedures, and what their participation would involve. They were also informed that they had the right to

withdraw from the study at any time, for any reason, without facing any negative consequences. This ensured that participants could make an informed decision about their involvement and maintain full control over their participation throughout the research process.

# 4 Results

## 5.1 Quantitative Data

This section presents the results of the t-tests comparing the ADHD and neurotypical groups on the NASA-TLX and USE questionnaire measures across the progressive disclosure and minimal data conditions. It also includes comparisons within the ADHD group and within the neurotypical group across both conditions.

An independent sample t-test data analysis was conducted to compare the means between the groups and conditions. The quantitative data was gathered through the NASA and USE questionnaires and the standard p-value threshold of 0.05 for statistical analysis was used for significance testing.

### 5.1.1 Comparison between the user groups

#### NASA TLX results: Progressive Disclosure

- ADHD group (N=5): M = 2.76, SD = 0.328
- Neurotypical group (N=5): M = 2.76, SD = 0.740

An independent samples t-test revealed no significant differences between the groups. The mean scores were identical, indicating an equal level of perceived workload.

#### USE questionnaire: Progressive Disclosure

The following table presents how both groups perceived the usability of the progressive disclosure prototype:

| Metric | ADHD Mean (SD) | Neurotypical Mean (SD) | p-value |
|---|---|---|---|
| **Usefulness** | 5.77 (0.95) | 5.08 (1.45) | 0.407 |
| **Ease of Use** | 6.40 (0.34) | 6.23 (0.70) | 0.647 |
| **Ease of Learning** | 6.90 (0.13) | 6.90 (0.13) | 1.000 |
| **Satisfaction** | 6.02 (1.08) | 5.62 (1.03) | 0.566 |

**Table 1:** Results from the independent samples t-test, comparing the mean of the results obtained using the USE scale for the “Progressive Disclosure” prototype.

In the progressive disclosure condition, the ADHD group reported slightly higher means in Usefulness, Ease of Use, and Satisfaction compared to Neurotypicals, though the differences did not reach statistical significance ($p>.05$)

NASA-TLX Results: Minimal Data

The NASA-TLX results for the "Minimal Data" condition (the condition where no extra information was available) indicate a difference between the two groups.

- ADHD Group (N=6): M= 3.90, SD= 0.901
- Neurotypical Group (N=8): M= 3.22, SD= 0.871

The independent samples t-test resulted in $t(12) = 1.414$, $p = 0.183$. While not statistically significant at the .05 level, the mean cognitive load for ADHD users was higher than for neurotypicals.

#### USE questionnaire: Minimal Data

The following table represents the usability metrics for the "Minimal Data" Prototype.

| Metric | ADHD Mean (SD) | Neurotypical Mean (SD) | p-value |
|---|---|---|---|
| **Usefulness** | 3.95 (0.85) | 4.51 (1.42) | 0.408 |
| **Ease of Use** | 5.55 (0.998) | 5.79 (0.706) | 0.249 |
| **Ease of Learning** | 6.50 (0.41) | 6.71 (0.38) | 0.332 |
| **Satisfaction** | 4.14 (1.00) | 4.80 (1.94) | 0.465 |

**Table 2:** Results from the independent samples t-test, comparing the mean of the results obtained using the USE scale for the "Minimal Data" prototype.

The ADHD group reported lower means across every metric compared to the neurotypical group. Notably, the ADHD group's satisfaction (4.14) and usefulness (3.95) ratings were considerably lower than those in the Progressive Disclosure condition.

### 5.1.2 Comparison within the ADHD user group

#### NASA TLX results: Progressive disclosure vs minimal for the ADHD group

This section compares the performance of ADHD participants with the "progressive disclosure" prototype vs the "minimal data" prototype.

There was a significant difference in the cognitive load experienced by the ADHD users depending on the interface design:

- ADHD with progressive disclosure: M= 2.76, SD= 0.328

- ADHD with minimal data: M= 3.90, SD= 0.901

The t-test showed t(9)=2.665, p=0.026(two-sided). This indicates a statistically significant increase in cognitive load when the ADHD group used the "minimal data" prototype compared to the one with progressive disclosure (containing extra data).

### USE Questionnaire: Progressive disclosure vs minimal data for the ADHD group

The ADHD group's evaluation of the two prototypes showed a clear preference for the interface providing more data (tags, duration, etc):

| Metric | Extra Data (Mean) | Minimal Data Mean (SD) | p-value(two-sided) |
|---|---|---|---|
| **Usefulness** | 5.77 (0.95) | 3.95 (0.85) | 0.009 |
| **Ease of Use** | 6.40 (0.34) | 5.55 (0.99) | 0.107 |
| **Ease of Learning** | 6.90 (0.13) | 6.50 (0.41) | 0.073 |
| **Satisfaction** | 6.02 (1.08) | 4.14 (1.00) | 0.015 |

**Table 3:** Results from the independent samples t-test, comparing the mean of the results obtained using the USE scale for ADHD participants.

The ADHD group was significantly more satisfied and found the system significantly more useful when the extra data was available.

### NASA TLX results: Neurotypicals with progressive disclosure vs minimal data

The cognitive load for neurotypical users showed a slight increase when data was removed, though the change was not statistically significant:

- Neurotypicals with extra data: M= 2.76, SD= 0.328
- Neurotypicals with minimal data: M= 3.22, SD= 0.871
- Significance: t(11) = -1.128, p= 0.283

### USE Questionnaire: Neurotypicals with progressive disclosure vs minimal data

The neurotypical group's evaluation of the two prototypes indicates a general preference for more data, though the margins were smaller than those seen in the ADHD group:

| Metric | Extra Data (Mean) | Minimal Data | p-value(two-sided) |
|---|---|---|---|

| | | Mean (SD) | |
|---|---|---|---|
| **Usefulness** | 5.77 (0.95) | 4.51(1.42) | 0.112 |
| **Ease of Use** | 6.40 (0.34) | 5.79 (0.70) | 0.103 |
| **Ease of Learning** | 6.90 (0.13) | 6.71 (0.38) | 0.342 |
| **Satisfaction** | 6.02 (1.08) | 4.80 (1.94) | 0.229 |

**Table 4:** Results from the independent samples t-test, comparing the mean of the results obtained using the USE scale for neurotypical participants.

### Summary of the Quantitative Results

There were differences in cognitive load and user satisfaction for all users between prototypes, but they were higher for the ADHD group. Both groups showed an increased satisfaction and a decrease in cognitive load when having more data to browse through.

#### Cognitive Load

For the "progressive disclosure" prototype, based on the data collected using the NASA-TLX scale, both neurotypicals and ADHD users had the same mean in cognitive load, but for the "minimal data" prototype, there was a difference of 0.675 (Neurotypicals Mean = 3.225, ADHD Mean = 3.9) between their means, indicating that the ADHD group found it to require more mental resources.

When comparing the neurotypical groups (with and without progressive disclosure), findings show that having more data at their disposal helped to decrease cognitive load slightly. However, the ADHD groups showed a statistical significance when the prototypes were compared against each other; the group with more data experienced significantly lower cognitive load when completing the task ("progressive disclosure" prototype mean = 2.76, "minimal data" prototype mean = 3.90).

#### User Satisfaction

Similar to the cognitive load, while the neurotypicals were also more satisfied with the extra data, the ADHD group was significantly more satisfied with it than with the version lacking it.

## 5.2 Qualitative Data

Following the guidelines proposed by Braun and Clarke (2006), a thematic analysis was performed on the qualitative data obtained. First, the data was transcribed using the recordings of each interview, then read through and filtered to only include relevant information to the study's research questions.

Initial codes were created to organize the data systematically and then themes were extracted from it. Each theme was defined and named accordingly, with examples to support it.

**The themes were divided into two sections: neurotypicals and ADHD. By comparing them, we aimed to understand the differences between the user groups.5.2.1 Neurotypical Group**

The main themes identified in the study, and stated as the deciding factor when it came to making choices were: cover, tags, description and reviews. In most cases, the neurotypicals stated that the "cover" was the deciding factor, while the "tags", "description" and "reviews" were factors that solidified their decision.

When faced with minimal data, neurotypicals mostly relied on visuals and descriptions. One participant stated "I picked them based on the cover. All the other ones melted into each other" (P1, "minimal data" prototype) as a reason for picking a specific movie to watch at their own leisure. However, when asked to pick a movie for an event they have relied on the description, noting that "I read the description and it said family routine" to make their choice.

In the case of the "progressive disclosure" prototype, some users only used the covers to decide "The covers caught my attention. This movie looks like something I'd watch" (P3, "progressive disclosure" prototype), or used a combination of both covers and description "Some of them were interesting but I really like romance movies and history so when I saw the description it was more romance and then the picture more historical- I picked it. So essentially the description convinced me to pick it in the end." (P1, "progressive disclosure" prototype).

When faced with extra data, participants rarely felt the need to use it, and stated that there was enough data, or more than they required, as expressed by a participant "I did not really read the descriptions, I picked them based on the image. Did not need the extra data." (P2, "progressive disclosure" prototype).

However, when the data was non-existent, a few participants have expressed a desire to have it, as stated by a participant "I would want more data. I always like to check when it was made, the year. Also ratings were missing and maybe how many people have watched it, like a "views" type of feature." (P4, "minimal" prototype).

| **Theme** | **Description** | **Quotes** |
|---|---|---|
| Cover | Participants mentioned that the decisive factor in their choice was the movie cover. | "The covers caught my attention. This movie looks like something I'd watch" (P3, "progressive disclosure" prototype) |

| | | |
|---|---|---|
| Tags | Participants stated that their decision was made after reading the tags and made sure their assumptions were correct. | “The cover wasn’t telling me a lot so I checked the description and the tags.” (P1, “progressive disclosure” prototype) |
| Description | Participants consistently checked the description before making their choices. | “The description made me more interested.” (P4, “minimal data” prototype) |
| Reviews | Participants mentioned that the reviews influenced their choice, or would have when it was missing. | “What mattered for me is the description, what kind of movie it was, and what it is about and maybe more reviews.. ”(P1, “progressive disclosure” prototype),<br>“I wanted to see reviews, like a score.”(P2, “minimal data” prototype) |
| Enough Data | Participants consistently stated that the data presented was enough. | “I think it was enough data. Usually the description and the tags do it for me” (P3, “progressive disclosure” prototype),<br>“I think the data was enough for me. Less is more you know. For me personally it is enough.I don’t need more.” (P2, “minimal data” prototype) |
| Not enough data | Participants mentioned the need for more data when making a decision. | “I think ratings were missing. I mostly look at ratings and title when I pick movies so that was the only thing missing for me.”(P5, “minimal data” prototype),<br>“I want the banner like the one in Netflix where they show a trailer or a scene out of the movie. I think the poster and the banner could be misleading.” (P8, “minimal data” prototype) |

**Table 5:** Themes resulting from the thematic analysis on the neurotypical groups.

#### 5.2.2 ADHD Group

The main themes identified in the study and stated as factors in the decision making of the ADHD group were “Cover”, “Tags”, “Genre”, “Ratings”, “Running time”,

“Description”. Participants used a wide range of factors, with most of the individuals showcasing a preference for a specific movie detail.

In the case of the “progressive disclosure” prototype, participants made their decision taking the extra data into account, for both tasks given to them.

One participant stated that “I did take the extra data into consideration - the running time” (P4, “progressive disclosure” prototype), while another stated “I picked this one because the premise is weird. What is a world of shifting cubes? But now that I see its animation I might not, but I would still probably be intrigued... And also this one, because it's Sci-Fi dystopian and has good ratings” (P1, “progressive disclosure” prototype). In a similar fashion, most of the participants made a decision after checking a specific detail, rarely picking based on the cover alone.

Moreover, when asked if there was enough data to make a decision, participants consistently stated that they would want more, with one participant stating that “I look at the cast and movie director usually so I would like to have that” (P5, “progressive disclosure” prototype). At the same time, a minority stated that it was enough; “I think there was enough data” (P1, “progressive disclosure” prototype).

The theme titled “No need for progressive disclosure” emerged after several participants have mentioned a dislike for the design strategy. One participant in particular has grown frustrated when using it, stating that “The details should be necessary to have, not hidden” (P4, “progressive disclosure” prototype).

The situation changed in the case of the “minimal data” prototype, as the participants were forced to make a decision based on minimal data. The decisions were mostly made using both the covers and the description of the movie “I picked it because I really like the poster. I locked it in using the description” (P7, “minimal data” prototype). The reasoning sometimes went further than the presented data, as some individuals stated that they inferred the genre of the movie based on the description: “It was the cover but that also introduced the genre and I was more interested in that. I inferred the genre” (P6, “minimal data” prototype).

Some of the participants had difficulties when making a decision and devised their own strategies, such as making a top after reading the movie descriptions: “I selected 3... The first one would be the star, because I like fantasy. It’s the most aligned with my preferences” (P6, “minimal data” prototype).

Differently from the neurotypical group, the ADHD user group counterpart consistently expressed a need for more data in their decision making process.

| **Theme** | **Description** | **Quotes** |
|---|---|---|
| Cover | Participants mentioned that the | “I’m judging them by the |

| | | |
|---|---|---|
| | decisive factor in their choice was the movie cover. | covers” (P4, “progressive disclosure” prototype) |
| Tags | Participants stated that their decision was made after reading the tags and made sure their assumptions were correct. | “I saw old fashioned and I went for it” (P5, “progressive disclosure” prototype) |
| Genre | Participants consistently checked the genre before making their choices. | “I went more after the genre than the cover and looks” (P3, “progressive disclosure” prototype) |
| Ratings | Participants made their decision based on the user ratings presented. | “Because it has good ratings”(P1, “progressive disclosure” prototype) |
| Running Time | Participants took into consideration the running time of the movie. | “I did take the extra data into consideration - the running time”(P4, “progressive disclosure” prototype) |
| Description | Participants checked the description to make their decision. | “First I saw the title and then the cover, and then I had to read what it was about. In the end I picked it because of the description” (P3, “minimal data” prototype) |
| Need more data | Participants stated that they would be more satisfied with being presented extra data regarding the movie when making their choice. | “Reviews and actors were missing for me, and the movie duration. Definitely more data would be better.” (P2, “minimal data” prototype),<br>“Maybe have the actors so I could avoid problematic actors like Kevin Spacey, because he is a bad person” (P3, “progressive disclosure” prototype),<br>“I would like to see the languages and subtitles” (P4, “progressive disclosure” prototype) |
| No need for | Participants expressed a negative | “I don’t see the point of |

| | | |
|---|---|---|
| progressive disclosure | opinion towards the way the data was presented using the progressive disclosure design strategy. | having an extra button to show it." (P1, "progressive disclosure" prototype), "The data should be necessary to have, not hidden" (P3, "progressive disclosure" prototype) |

**Table 6:** Themes resulted from the thematic analysis on the ADHD groups.

# 5 Discussion

## 5.1 Result discussion

The purpose of this study was to investigate how data visualization within movie recommender systems affects the cognitive load and satisfaction of ADHD users compared to neurotypical users. To fulfill this purpose, the study reflects on the following findings:

**RQ1: How does data visualization affect the cognitive load of users with ADHD in recommender systems?**

In our theoretical framework, the design strategy identified as a primary tool for reducing cognitive load was progressive disclosure (Yabloski, 2020). The theoretical goal was to "streamline" the interface by hiding secondary details like tags and duration until they were needed. "Sensory noise" was expected to be prevented (Wu et al., 2021) and help ADHD users stay focused on the primary task of picking a movie by limiting the extraneous load (Sweller et al., 2011). By providing visual anchors, the interface allowed users to avoid the heavy lifting of sustained reading, resulting in a statistically significant drop in the cognitive load ($p = .026$).

However, the qualitative findings revealed that ADHD participants expressed the need for the additional information to be visible at all times rather hidden behind progressive disclosure. This creates an interesting contradiction between the quantitative and qualitative findings. While cognitive load was reduced, participants did not necessarily attribute this benefit to the progressive disclosure mechanism itself. Instead, many participants emphasized that the additional information was essential for their decision-making process and expressed frustration when it was not immediately accessible. Thai suggests that the reduction in cognitive load maybe attributed more to the availability of relevant information than to progressive disclosure as a design strategy. Therefore, the issue may not only be the amount of information presented, but also how that information is organized and revealed to the

user. ADHD users may benefit from interfaces that provide access to rich contextual information without requiring additional interactions to uncover it.

**RQ2: Are there differences between ADHD and neurotypical users in how data visualization influences cognitive load?**

While both groups perceived the version containing extra data as more useful, this preference was considerably more pronounced among participants with ADHD. With an absence of tags, duration of the movie details, and genre, participants were required to rely heavily on the movie covers, titles and summaries when making a choice. This lack of certainty made it harder to pick a movie, leading to the higher NASA-TLX scores and lower satisfaction for the ADHD group. In contrast, the neurotypical participants remained relatively unaffected by the change of layout. Although this group also had a preference for the version with the more data, they weren't as dissatisfied as the ADHD group.

**RQ3: How does data visualization influence user satisfaction among users with ADHD?**

While both groups expressed a preference for the "extra data" prototype, the ADHD group found the detailed visualization significantly more useful than the neurotypical group. The extra data served as a shortcut, allowing ADHD users to bypass the cognitively exhausting process of analyzing descriptive text in favor of making quick, data-driven decisions.

Moreover, while the neurotypical participants were generally satisfied with the standard amount of data provided in the progressive disclosure prototype, the ADHD group expressed a desire to always have more data available, without the additional interaction cost of progressive disclosure. available, without the additional interaction cost of progressive disclosure.

## 5.2 Method discussion

One of the weaknesses of our study is the small sample size, as we did not have access to a big pool of ADHD participants. This might have limited the power to achieve statistical significance on some aspects, such as cognitive load and overall satisfaction, when comparing the ADHD groups. A larger sample might have revealed more subtle differences.

Another weakness of our study was that the method was conducted in a mixed format, with some participants attending online and others in person due to availability issues. This difference in testing environments may have influenced the results, as participants could have experienced the sessions differently depending on the setting.

One more weakness is that the prototypes used during testing were simplified Figma designs rather than fully functional recommender systems. While the prototype allowed us to isolate and evaluate specific design elements, they did not replicate the full experience of interacting with commercial streaming platforms such as Netflix or Spotify. However, this decision was made intentionally to reduce potential familiarity bias. Using existing platforms could have influenced participant choices based on prior experience with the interface, previous exposure to the recommended content, or established usage habits. By creating custom prototypes and using unfamiliar movie options, we were able to ensure that participants based their decisions primarily on the information presented within the interface rather than on external factors. Nevertheless, the simplicity of the prototypes may limit the generalization of the results to real-world recommender systems, where users interact with may more complex and personalized environments.

One of the key strengths of our study was the focus on specifically on people with ADHD rather than relying on assumptions about how users in general might respond. This allowed us to gain more meaningful insights into a group that is often underrepresented in traditional UX research.

Validity was prioritized by selecting frameworks like the NASA-TLX and USE, which are well-suited for capturing the subtle differences in neurodivergent user experiences. To support reliability, the study followed a consistent procedure where all participants were given the same instructions and completed the same tasks, helping to ensure that the study could be replicated and produce similar results.

To further strengthen the findings, data triangulation was applied (Carter et al., 2014). By combining quantitative measures with qualitative semi-structured interviews, the study was able to go beyond the numbers and compare them with participants' own descriptions of their experiences. This helped ensure that our conclusions were not only supported by data but also reflected the real experience of the users.

Overall, we believe that the purpose of this study has been successfully fulfilled. Our goal was to see if data visualization could lower the cognitive load on ADHD users, and our method comparing two different prototypes provided a clear answer.

# 6 Conclusions and further research

## 6.1 Conclusions

In this paper, our findings indicate that ADHD users may experience how data is presented differently than their neurotypical counterparts. When presented with more data, the cognitive loead decreased, and satisfaction rose significantly, which indicates a strong desire for more information being made available.

To achieve these insights, our study investigated how data visualization and progressive disclosure influence cognitive load and user satisfaction in a movie streaming website using a recommender system. The usability testing was conducted with both ADHD and neurotypical participants, comparing an interface with no extra data against a prototype that visualized extra data (such as tags, genre, duration). Participants completed a series of tasks where they had to choose unfamiliar movies, specifically designed for the study to minimize bias, and the study measured their experience using the NASA-TLX and USE standardized scales along-side follow-up semi-structured interviews.

Our results revealed that while both groups preferred the progressive disclosure prototype version more, the ADHD group demonstrated higher levels in satisfaction. Furthermore, unlike the neurotypical participants, who were generally comfortable with the interface, participants with ADHD consistently expressed a need for more information across both prototypes.

These findings indicate that individuals with ADHD need extra visual aids when making choices online. Having access to extra data acts as a necessity aid that makes decision-making much easier for them and directly reduces their cognitive load.

Ultimately, these results show that standard UX guidelines are missing an important piece of the accessibility puzzle by assuming that merely a clean or minimalistic interface works for everyone. Real inclusivity in design means moving beyond a one-size-fits-all approach. The focus should be shifted towards providing useful visuals and move away from treating ADHD as “a problem to be fixed”. Instead, giving users access to the data they need protects their independence and gives them the power to make their own choices comfortably.

## 6.2 Further research

This study offers a preliminary base for further studies to utilize. While the findings indicate that the ADHD users do require more data when faced with choices, it has not touched upon the influence of the data display, as only progressive disclosure was used as a design strategy. Therefore, future research could investigate how ADHD users react to different design choices in the context of web interfaces.

To build on our findings, future research could employ eye-tracking technology to see exactly what ADHD users experience visual friction, or use a strict, single-environment set-up to eliminate the environmental “noise” encountered.

Additionally, future research could look into how design choices play out in completely different environments, like e-commerce. Shopping online often involves massive information clutter, reviews and product specifications, for instance. Future

work could investigate how ADHD users experience how such platforms display information, both progressively and by default.

# 7 References


Anne‐Laure Le Cunff, Dommett, E., & Giampietro, V. (2023). Neurophysiological measures and correlates of cognitive load in attention‐deficit/hyperactivity disorder (ADHD), autism spectrum disorder (ASD) and dyslexia: A scoping review and research recommendations. *European Journal of Neuroscience*, *59*(2), 256–282. https://doi.org/10.1111/ejn.16201

American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. 5th ed., Arlington, VA, American Psychiatric Publishing, 2013.

Belabbes, M. A., Ruthven, I., Moshfeghi, Y., & Rasmussen Pennington, D. (2023). Information overload: A concept analysis. *Journal of Documentation*, *79*(1), 144–159. https://doi.org/10.1108/JD-06-2021-0118

Braun, Virginia, and Victoria Clarke. "Using Thematic Analysis in Psychology." *Qualitative Research in Psychology*, vol. 3, no. 2, 2006, pp. 77–101, https://doi.org/10.1191/1478088706qp063oa.

Britto, T. C. P., & Pizzolato, E. B. (2016). Towards Web Accessibility Guidelines of Interaction and Interface Design for People with Autism Spectrum Disorder. *Advances in Computer-Human Interaction*, 138–144. https://www.thinkmind.org/download.php?articleid=achi_2016_7_10_20374

Carter, N., Bryant-Lukosius, D., DiCenso, A., Blythe, J., & Neville, A. J. (2014). The use of triangulation in qualitative research. *Oncology nursing forum*, 41(5), 545–547. https://doi.org/10.1188/14.ONF.545-547

Chattopadhyay, R. (2025). Inclusive digital training systems and learning equity in the workplace: evidence from a neurodiverse workforce. *Journal of Workplace Learning*, 1–17. https://doi.org/10.1108/jwl-06-2025-0178

Cao, A., Chintamani, K. K., Pandya, A. K., & Ellis, R. D. (2009). NASA TLX:

Software for assessing subjective mental workload. *Behavior Research Methods*, *41*(1), 113–117. https://doi.org/10.3758/brm.41.1.113

Dai, J., Miedema, J., Hernandez, S., Sutton-Lalani, A., & Moffatt, K. (2023). *Cognitive Accessibility of Digital Payments: A Literature Review*. https://doi.org/10.1145/3587281.3587294

Ding, G.-J., Hwang, T. K. P., & Kuo, P.-C. (2020). Progressive Disclosure Options for Improving Choice Overload on Home Screen. *Advances in Intelligent Systems and Computing*, 173–180. https://doi.org/10.1007/978-3-030-51828-8_23

Ekstrand, M. D., & Willemsen, M. C. (2016). *Behaviorism is Not Enough*. https://doi.org/10.1145/2959100.2959179

Iyengar, S. S., & Lepper, M. R. (2000). When choice is demotivating: Can one desire too much of a good thing? *Journal of Personality and Social Psychology*, *79*(6), 995–1006. https://doi.org/10.1037/0022-3514.79.6.995

Kalantari, N., Bagale, A., & Motti, V. G. (2025). Designing assistive technologies for and with neurodivergent users: considerations from research practice. *Interacting with Computers*. https://doi.org/10.1093/iwc/iwaf037

Khan, S., Knijnenburg, B., & Dixon, E. (2024). *Evaluating ADHD Users' Experience with Recommender Systems*. https://doi.org/10.1145/3640544.3645222

Kunkel, J., Loepp, B., & Ziegler, J. (2017). A 3D Item Space Visualization for Presenting and Manipulating User Preferences in Collaborative Filtering. *Proceedings of the 22nd International Conference on Intelligent User Interfaces*, 3–15. https://doi.org/10.1145/3025171.3025189

Lund, A. (2001, January). *Measuring Usability with the USE Questionnaire*. ResearchGate. https://www.researchgate.net/publication/230786746_Measuring_Usability_with_the_USE_Questionnaire

Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, *101*(2), 343–352. https://doi.org/10.1037/0033-295x.101.2.343

Moroyoqui, M., Espina, D., Osuna, H., & Escobedo, L. (2022). *SmarTasko: Supporting short and spontaneous activities of daily living of ADHD individuals*. *43*, 1–6. https://doi.org/10.1145/3565494.3565500

Pfitzner, D. (2003). A unified taxonomic framework for information visualization. *In APVis'03: Proceedings of the Asia-Pacific Symposium on Information Visualisation*, *24*. https://doi.org/10.5555/857080.857087

Ribeiro, D. M., Melo, F. V., Negromonte, V., Matias, G. W., Farias, A., Azul, C., Chaves, A. P., & Gama, K. (2024). A Comparative Study on Accessibility for Autistic Individuals with Urban Mobility Apps. *ArXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2410.14033

Richthammer, C., Sänger, J., & Pernul, G. (2017). Interactive Visualization of Recommender Systems Data. *Proceedings of the 4th Workshop on Security in Highly Connected IT Systems*, 19–24. https://doi.org/10.1145/3099012.3099014

Sawyer, Steven. "Analysis of Variance: The Fundamental Concepts." ResearchGate, Maney Publishing, Apr. 2009, www.researchgate.net/publication/272311020_Analysis_of_Variance_The_Fundamental_Concepts.

Sieghart, S., Björn Rohles, & Bessemans, A. (2024). *Empowering Independence Through Design: Investigating Standard Digital Design Patterns For Easy-to-Read Users.* 1–18. https://doi.org/10.1145/3613904.3641911

Spiel, K., Hornecker, E., Williams, R. M., & Good, J. (2022). ADHD and technology research – Investigated by neurodivergent readers. *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, Article 547, 1–21. https://doi.org/10.1145/3491102.3517592

Sweller, J., Ayres, P., & Kalyuga, S. (2011). *Cognitive Load Theory*. Springer New York. https://doi.org/10.1007/978-1-4419-8126-4

Tang, K., Chen, K., Lin, Q., Li, Y., & Jiang, Z. (2025). Designing a Gamified Driving Mode with Music and Mapping to Assist Drivers with ADHD on Monotonous Road Segments. *Adjunct Proceedings of the 17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 250–255. https://doi.org/10.1145/3744335.3758513

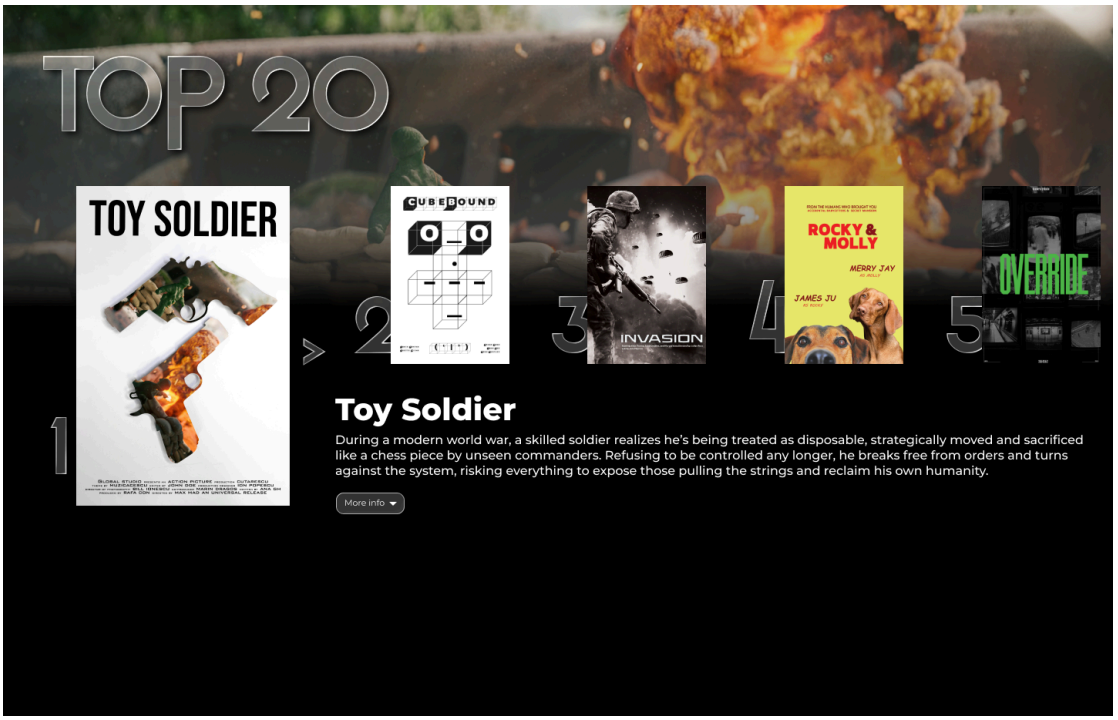
Wu, K., Petersen, E., Ahmad, T., Burlinson, D., Tanis, S., & Szafir, D. A. (2021). Understanding Data Accessibility for People with Intellectual and Developmental Disabilities. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/3411764.3445743

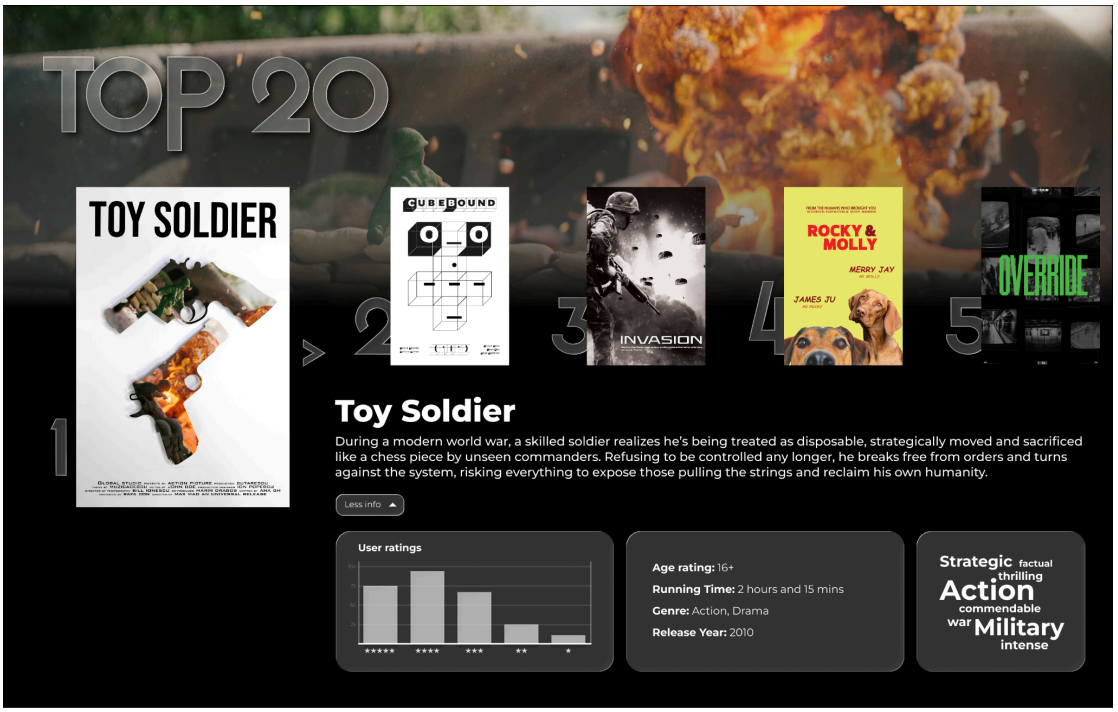
Yablonski, J. (2020). Laws of UX: Using Psychology to Design Better Products and Services: Using Psychology to Design Better Products and Services. O'Reilly Media, Incorporated.

# 8 Appendixes

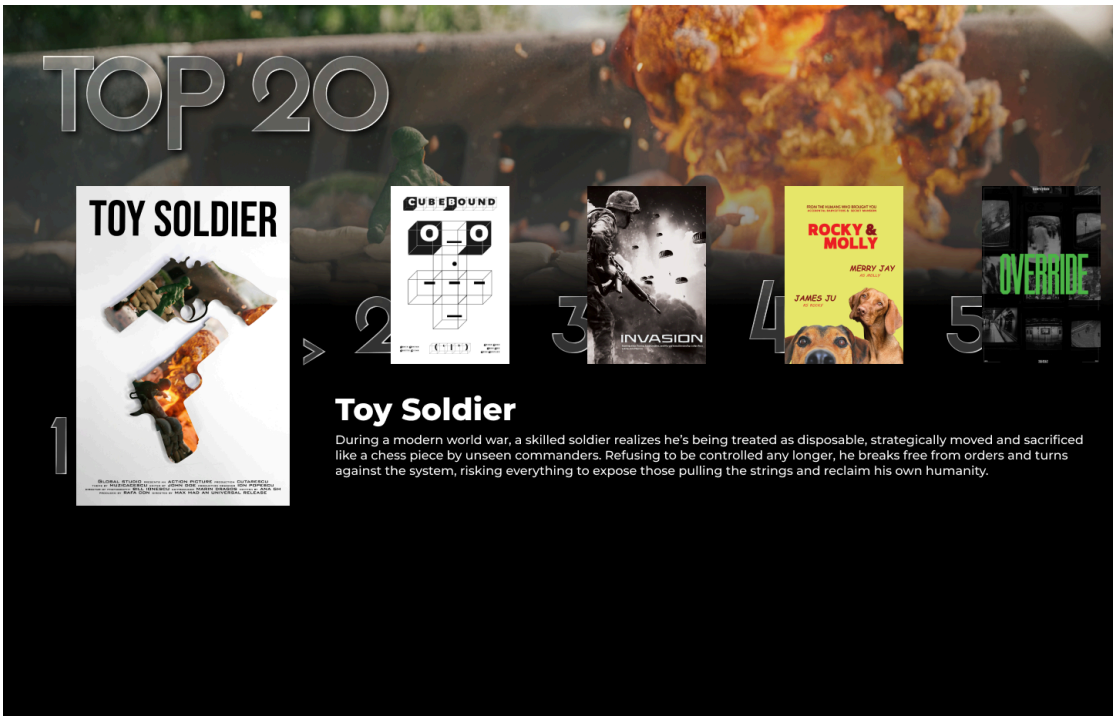
## Appendix 1 - Prototype

With progressive disclosure:




Without progressive disclosure:




Link to the figma prototype: https://www.figma.com/proto/RM5j5izGCaUL29bKknAuMU/Top_20_Prototype?node-id=1-2&p=f&t=7OsJlIcprkfBAckV-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2

## Appendix 2 - Questionnaire

The questionnaire is divided into two parts, consisting of NASA-TLX and USE.

Section one is meant to measure the cognitive load of the users, using the NASA-TLX template.

*Cognitive Load*

The following questions ask about the workload you experienced while completing the tasks using the recommender system.

Please rate each statement based on how demanding the task felt during the interaction

| How mentally demanding was the task? | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | |
| Very Low | o | o | o | o | o | o | o | Very High |

| How hurried or rushed was the pace of the task? | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | |
| Very Low | o | o | o | o | o | o | o | Very High |

| How successful were you in accomplishing what you were asked to do? | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | |
| Very Low | o | o | o | o | o | o | o | Very High |

| How hard did you have to work to accomplish your level of performance? | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | |
| Very Low | o | o | o | o | o | o | o | Very High |

| How insecure, discouraged, irritated, stressed, and annoyed were you when executing the task? | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | |
| Very Low | o | o | o | o | o | o | o | Very High |

Section two is meant to measure the user satisfaction using the USE template.

*User Satisfaction*

Indicate how much you agree or disagree with the following statement based on your experience during the usability test.

Use the scale provided to select the option that best represents your opinion.

| **Usefulness** | Strongly disagree | Disagree | Slightly disagree | Neutral | Slightly agree | Agree | Strongly agree |
|---|---|---|---|---|---|---|---|
| The extra data helped me be more effective when accomplishing the | | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| task. | | | | | | | |
| The extra data helped me be more productive. | | | | | | | |
| The extra data was useful when picking a movie. | | | | | | | |
| The extra data gives me more control over the activities in my life | | | | | | | |
| The extra data saves me time when I use it | | | | | | | |
| The extra data meets my needs | | | | | | | |
| The extra data does everything I would expect it to do. | | | | | | | |

| **Easy to Use** | Strongly disagree | Disagree | Slightly disagree | Neutral | Slightly agree | Agree | Strongly agree |
|---|---|---|---|---|---|---|---|
| It is easy to use | | | | | | | |
| It requires the fewest steps possible to accomplish what I want to do with it | | | | | | | |
| Using it is effortless | | | | | | | |
| I can use it without written instructions | | | | | | | |
| Both occasional and regular users would like it | | | | | | | |
| I can use it successfully every time | | | | | | | |

| **Ease of Learning** | Strongly disagree | Disagree | Slightly disagree | Neutral | Slightly agree | Agree | Strongly agree |
|---|---|---|---|---|---|---|---|
| I learned to use it quickly | | | | | | | |
| I easily remember how to use it | | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| It is easy to learn to use it | | | | | | | |
| I quickly became skillful with it | | | | | | | |

| **Satisfaction** | Strongly disagree | Disagree | Slightly disagree | Neutral | Slightly agree | Agree | Strongly agree |
|---|---|---|---|---|---|---|---|
| I am satisfied with it | | | | | | | |
| I would recommend it to a friend | | | | | | | |
| It is fun to use | | | | | | | |
| It works the way I want it to work | | | | | | | |
| I feel I need to have it | | | | | | | |
| It is pleasant to use | | | | | | | |

## Appendix 3 - Semi-structured Interview Questions

1. Why did you pick this specific movie to watch?
2. What made you pick the movie?
3. Was the user interface helpful when picking a movie?
4. Was the information useful / would you need more data? Less data

We will keep the questions open-ended and try to obtain a follow-up response.

## Appendix 4 - Informed Consent

*Purpose of the Study*

You are hereby invited to take part in a research study. The research team conducts the study as part of their bachelor's thesis in Informatics at Jönköping University. The purpose is to investigate the effects of data visualization on users with ADHD.

*Purpose of the Study*

In this study, you will perform a usability test on a Figma prototype. A set of scenarios will be introduced. Consequently, you will answer a questionnaire based on your experience. Finally, a short follow-up semi-structured interview will be

conducted. The session is expected to take around 20 minutes. You will be asked for consent to record both your screen and voice during all parts of this study. Your participation in this study is completely voluntary.

*Purpose of the Study*

The data collected during the study is for academic purposes only. No private information will be asked for or disclosed in the research report. All data will remain anonymized. The information will be securely stored during the research period of three months and then deleted immediately. The researchers will make sure no harm is inflicted on you during your study.

*Participation Right*

You, as a participant, have the right to withdraw from the study at any given time during or after any part of the study without providing a reason or without any negative consequences.

*Right and Benefits*

You will receive a _____ from the researchers as an expression of gratitudine for contributing to the study. Your contribution to the research will expand knowledge in the field of Informatics and encourage further research and development in this area.

If you have any questions or want to withdraw your data from the study at any time during the research period, you can contact us, and your data will be eliminated immediately.

*Contact Information*


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*Consent*

By signing below, you confirm that:

- You are 21 years or above
- You have read and understood the information and conditions
- You give full consent to your participation in the study

Participant Name: _____ Researcher Name:_____

Date: _____ Date: _____

Signature: _____ Signature: _____

## Appendix 5 - Detailed overview of the literature review

| Citation | Method | Neurodivergencies studied | Summary |
|---|---|---|---|
| Anne‐Laure Le Cunff, Dommett, E., & Giampietro, V. (2023). Neurophysiological measures and correlates of cognitive load in attention‐deficit/hyperactivity disorder (ADHD), autism spectrum disorder (ASD) and dyslexia: A scoping review and research recommendations | Literature Review | ADHD, ASD and dyslexia | A literature review that analyzes different ways to measure cognitive load in people with ADHD, ASD and dyslexia. Among them, EEG and fMRI are the most used and despite their popularity in studies, eye tracking, fNIRS and MEG are underutilised in the context of neurodivergent individuals. There are several theories on why that is, but it is stated that they are still useful and are a valuable way of measuring cognitive load. |
| Chattopadhyay, R. (2025). Inclusive digital training systems and learning equity in the workplace: evidence from a neurodiverse workforce. | Quantitative Study | Studied as a collective | A study aiming to investigate how the inclusive design features help maintain the fairness and effectiveness for neurodivergent users in the context of digital training systems. Their results indicate that the design features significantly enhance equity between neurotypicals and neurodivergents and should be integrated into learning |

| Citation | Method | Neurodivergencies studied | Summary |
|---|---|---|---|
| | | | strategies. A future research direction they indicate is to understand how people with neurodivergent conditions interact with distinct inclusive features. |
| Dai, J., Miedema, J., Hernandez, S., Sutton-Lalani, A., & Moffatt, K. (2023). Cognitive Accessibility of Digital Payments: A Literature Review. | Literature review | Cognitive Decline and collective neurodivergencies | A study that highlights the need for inclusive features in online banking apps to accommodate people who experience cognitive decline or neurodivergents. |
| Ding, G.-J., Hwang, T. K. P., & Kuo, P.-C. (2020). Progressive Disclosure Options for Improving Choice Overload on Home Screen. | Quantitative Study | Cognitive Disabilities due to old age | A study that tested whether progressive disclosure helps reduce choice overload. The results point towards an improvement, especially for older people. |
| Ekstrand, M. D., & Willemsen, M. C. (2016). Behaviorism is Not Enough. | Provides guidelines | none | A paper that argues that user input is very valuable, with a focus on recommender systems. They argue that without properly understanding what the individuals want, or assuming their preferences will make the experience worse. |
| Farmer, G. D., Baron-Cohen, S., & Skylark, W. J. (2017). People With Autism Spectrum Conditions Make More Consistent Decisions | Qualitative and Quantitative Study | Autism Spectrum Conditions | A study that showcases how people with ASC (Autism Spectrum Conditions) make more consistent decisions and are less likely to change them afterwards, unlike their neurotypical counterparts. The researchers conducted a study with a |

| Citation | Method | Neurodivergencies studied | Summary |
|---|---|---|---|
| | | | basic riskless task, which revealed that people with autism could potentially avoid biases and are less likely to be influenced by decoys used in real-life marketing. |
| Kalantari, N., Bagale, A., & Motti, V. G. (2025). Designing assistive technologies for and with neurodivergent users: considerations from research practice. Interacting with Computers. | Literature Review | Collective neurodivergencies | A literature review that identifies the gaps in the design methodologies that studied neurodivergent individuals. They highlight the need of involving neurodivergents in the design to make sure their needs are met. |
| Khan, S., Knijnenburg, B., & Dixon, E. (2024). Evaluating ADHD Users' Experience with Recommender Systems. | Qualitative Study | ADHD | A study that highlights the fact that, in general, ADHD users are overlooked when designing technologies or even testing them. By studying the case of recommender systems and their effects on adults suffering from ADHD, the researchers found that the users are negatively affected by them. Most subjects have reported that the systems encourage bad behaviours, and the only effective way to counter them was to not engage with recommended items at all. |
| Kunkel, J., Loepp, B., & Ziegler, J. (2017). A 3D Item Space Visualization for Presenting and Manipulating User | Qualitative study | none | A study that builds upon a proposed set of interactive visualization guidelines and applies it to a 3D environment. The results show an increase in user satisfaction, but also point to the fact that |

| Citation | Method | Neurodivergencies studied | Summary |
|---|---|---|---|
| Preferences in Collaborative Filtering. | | | users are more likely to use it only when they are not sure what to pick. |
| Moroyoqui, M., Espina, D., Osuna, H., & Escobedo, L. (2022). SmarTasko: Supporting short and spontaneous activities of daily living of ADHD individuals. | Qualitative Study | ADHD | A study that proposes a new app that should make starting and finishing tasks for people suffering from ADHD easier. By taking into consideration the factors that make daily actions difficult and by reducing the cognitive load and stimuli, the researchers have created a low-fidelity prototype. |
| Ribeiro, D. M., Melo, F. V., Negromonte, V., Matias, G. W., Farias, A., Azul, C., Chaves, A. P., & Gama, K. (2024). A Comparative Study on Accessibility for Autistic Individuals with Urban Mobility Apps. ArXiv (Cornell University). | Provide guidelines | Autism | A comparative study that analyzes the most popular urban mobility apps in Portugal using the Interface Accessibility guide for Autism. Their findings show that even the highest-scoring app had around 79% of points in total, with the lowest one having around 44%. The results emphasise the need for apps that are used in our daily lives to implement accessibility features. |
| Richthammer, C., Sänger, J., & Pernul, G. (2017). Interactive Visualization of Recommender Systems Data. Proceedings of the 4th | Provide guidelines | none | A study that proposes a set of interactive visualization guidelines within recommender systems to enhance the quality of recommendations and their transparency. |
| Sieghart, S., Björn Rohles, & Bessemans, A. (2024). Empowering Independence Through Design: | Qualitative and Quantitative Study | Cognitive Impairment | A study that examines how users with cognitive impairments interact with interfaces. By asking the subjects to complete certain tasks on mobile and |

| Citation | Method | Neurodivergencies studied | Summary |
|---|---|---|---|
| Investigating Standard Digital Design Patterns For Easy-to-Read Users. | | | laptop, the researchers found that due to the user-friendly nature of smartphones, app interfaces are easier for them to use and understand. |
| Spiel, K., Hornecker, E., Williams, R. M., & Good, J. (2022). ADHD and technology research – Investigated by neurodivergent readers. | Literature review | ADHD | A literature review that critically analyzes different technologies designed for people with ADHD. Their results showcase that studies often overlook ADHD individuals involvment when designing technologies for them, and concentrate on "correcting" their behaviour rather than accommodating it. |
| Tang, K., Chen, K., Lin, Q., Li, Y., & Jiang, Z. (2025). Designing a Gamified Driving Mode with Music and Mapping to Assist Drivers with ADHD on Monotonous Road Segments. Adjunct | Qualitative | ADHD | A study that proposes a graphical user interface for ADHD drivers to keep them engaged and minimize the risk of accidents. The researchers analyzed the issues that are most common and likely to distract the driver and found a way to mitigate them. The result is a prototype that combines music, clear visualization, and the concept of gamification to create a safer way to drive for ADHD individuals. |
| Wu, K., Petersen, E., Ahmad, T., Burlinson, D., Tanis, S., & Szafir, D. A. (2021). Understanding Data Accessibility for People with Intellectual and Developmental Disabilities. | Qualitative and Quantitative Study | Intellectual and Developmental Disabilities | A study that set out to understand how people with Intellectual and Developmental Disabilities (IDD) perceive different types of graphs. By conducting a simple test, the researchers found out that while both neurotypicals and IDD individuals preferred the same type of graphs for specific situations, the difference it made for IDD people was |

| Citation | Method | Neurodivergencies studied | Summary |
|---|---|---|---|
| | | | significant. In the conclusion, the authors state that the findings from this study should be integrated into universal design, as they vastly improve the experience of the IDD individuals, as well as not compromising the experience of others. |
| Britto, T. C. P., & Pizzolato, E. B. (2016). Towards Web Accessibility Guidelines of Interaction and Interface Design for People with Autism Spectrum Disorder. | Literature review/ Provides guidelines | Autism | A literature study that has identified a set of 28 guidelines extracted from selected works about accessibility for people with Autism. |